

Implementation of the Objectives

1. **File Compression Model for PDF Documents**

The file compression model will be implemented by integrating a PDF optimization algorithm into the system during the upload process. Scanned examination papers will be processed automatically to reduce file size through image compression, resolution adjustment, and removal of redundant metadata before being stored in the cloud. This approach will ensure reduced storage usage and faster upload times while maintaining acceptable document quality.

2. **Cloud-Based Storage System for Examination Papers**

The cloud-based storage system will be developed using web technologies and a cloud storage platform to support secure scanning, uploading, retrieval, and long-term preservation of examination papers. The system will include structured file organization, metadata tagging, and search functionality to enable quick access to specific documents. Secure authentication will be implemented to ensure that only authorized users can store and retrieve files.

3. **Automated Alert Model for Examination Score Upload**

The automated alert model will be implemented by integrating a notification service within the system. After examination scores are recorded or when submission deadlines approach, the system will automatically send alerts to lecturers via email, SMS or in-system notifications, reminding them to upload examination scores within the required timeframe.

4. **Access Control Model with Time-Based Permissions**

The access control model will be implemented using role-based and time-based authorization mechanisms. Administrators and lecturers will be able to assign access rights to specific users and define the duration for which access is granted. Once the specified time period expires, the system will automatically revoke access, ensuring controlled and secure use of sensitive academic documents.